

# Relational Grounding Failures in Zero-Shot Cross-Lingual Dialogue Summarization with Small Language Models

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## Abstract

We diagnose relational grounding failures in zero-shot English-to-Chinese dialogue summarization with small language models (SLMs). We develop and apply a Dialogue-Semantic Grounding Error Taxonomy to evaluate three open-weight SLMs alongside a supervised mBART baseline. The taxonomy captures errors in entities, events, speaker roles, circumstances, modality, discourse outcomes, omissions, and language or format issues. Our analysis shows that SLM hallucinations often involve relation-sensitive distortions of dialogue-relevant content: models may preserve source-relevant participants, events, or circumstances while altering who did what, under what conditions, or with what outcome. These findings position the taxonomy as a diagnostic evaluation framework for dialogue summarization, revealing relation-sensitive faithfulness failures that reference-based metrics may obscure.

## 1 Introduction

Small language models (SLMs) are attractive for resource-constrained, industrial, and privacy-sensitive summarization because they are efficient and locally deployable (Abrougui et al., 2026; Xu et al., 2025; Philippy et al., 2025). In cross-lingual dialogue summarization, however, it remains unclear whether SLMs can generate target-language summaries that are both fluent and faithfully grounded in the source dialogue. We define faithfulness as source support for the summary’s entities, events, speaker roles, circumstances, modality, and discourse outcomes.

We study zero-shot English-to-Chinese dialogue summarization, where models must identify salient dialogue content in English and express it in Chinese (Wang et al., 2022b, 2023a). Prior work shows that cross-lingual summarization is sensitive to

dataset construction, translation artifacts, and evaluation design (Bhattacharjee et al., 2023; Wang et al., 2023b), while reference-based automatic metrics may obscure how outputs fail to preserve dialogue meaning.

Building on prior factual-error taxonomies for dialogue summarization (Tang et al., 2022; Wang et al., 2022a), we develop and apply a Dialogue-Semantic Grounding Error Taxonomy for source-grounded diagnosis of zero-shot English-to-Chinese summaries. The taxonomy organizes hallucination and faithfulness failures around entities, events, participant roles, circumstances, modality, discourse outcomes, omissions, and cross-lingual output-form issues. Using this taxonomy alongside ROUGE, BERTScore, and OmniScore, we compare three open-weight SLMs with a supervised mBART baseline and show that SLM hallucinations often involve relation-sensitive distortions: models may preserve source-relevant participants, events, or circumstances while misrepresenting who did what, under what conditions, or with what outcome.<sup>1</sup>

| Label    | Core annotation trigger                                                                          |
|----------|--------------------------------------------------------------------------------------------------|
| H-ENT    | Wrong or unsupported person, entity, name, or object.                                            |
| H-EVT    | Added, changed, or fabricated event/action.                                                      |
| H-ROLE   | Correct event but wrong actor, recipient, speaker, or role.                                      |
| H-CIRC   | Wrong time, place, number, quantity, condition, or schedule.                                     |
| H-MOD    | Changed negation, uncertainty, intention, obligation, or modal force.                            |
| H-DISC   | Misstated cause-effect, contrast, order, decision, agreement, conclusion, or outcome.            |
| OMIT     | Missing salient event, participant, reason, decision, or outcome.                                |
| LANG     | English leakage, malformed Chinese, repetition, translation/format issue, or non-summary output. |
| NO-ERROR | No grounding, omission, language, format, or instruction-following error.                        |

Table 1: Dialogue-Semantic Grounding Error Taxonomy. H-\* labels indicate hallucination; OMIT and LANG capture missing content and output-form failures.

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<sup>1</sup>Project repository: <https://github.com/yunlee919/relational-grounding-dialogue-summarization>

## 2 Experimental Setup

### 2.1 Data

We evaluate models on XSAMSum, a cross-lingual dialogue summarization dataset from the ClidSum benchmark (Wang et al., 2022b), which extends the English SAMSum corpus (Gliwa et al., 2019) with professional summaries in multiple target languages. We focus on English-to-Chinese summarization, where models generate Chinese summaries from English messenger-style dialogues. For diagnostic evaluation, we use a fixed random sample of 100 test examples from the XSAMSum test split selected with seed 42. Dataset and evaluation-subset statistics are reported in Appendix A.1.

### 2.2 Models

We compare a supervised mBART-large-50 baseline (Liu et al., 2020; Tang et al., 2021) with three open-weight instruction-tuned SLMs under 10B parameters: Aya Expanse 8B (Dang et al., 2024), Gemma 4 E4B (Gemma Team, 2026), and Qwen 3.5 9B (Qwen Team, 2026). We selected these models to represent contemporary open-weight, instruction-tuned SLMs under 10B parameters that support multilingual text generation. The mBART baseline is fine-tuned on the XSAMSum training data, while the SLMs are evaluated in a zero-shot direct setting. Given an English dialogue, each SLM is prompted to generate a concise Simplified Chinese summary. The same prompt is used for all SLMs. Full model configurations and the shared prompt template are provided in Appendices A.2 and A.3, respectively.

### 2.3 Evaluation

We evaluate model outputs using reference-based ROUGE and BERTScore, the learned reference-based metric OmniScore, and dialogue-semantic grounding error analysis.

**Dialogue-semantic grounding error analysis.** To diagnose failures not captured by aggregate scores, we develop and apply the Dialogue-Semantic Grounding Error Taxonomy in Table 1. The taxonomy was induced from manual inspection of preliminary SLM summaries, where many errors preserved dialogue-relevant content but altered relations among speakers, events, circumstances, modality, or outcomes. We relate the categories to prior factuality typologies for abstractive and dialogue summarization (Pagnoni et al., 2021; Tang

et al., 2022; Wang et al., 2022a), but organize them around the dialogue-semantic component affected in a cross-lingual source-to-summary setting. In particular, the taxonomy separates entity, event, and participant-role errors, captures speaker-role misattribution through H-ROLE, discourse-relation distortions through H-DISC, omissions through OMIT, and target-language or output-form failures through LANG; Appendix A.8 provides an approximate comparison with prior taxonomies. These choices are also motivated by work on speaker-action structure, discourse relations, coreference, and context carryover (Chen and Yang, 2021; Aghajanyan et al., 2020). Summaries may receive multiple labels for independent error types.

**Reference-based metrics.** We compute ROUGE (Lin, 2004) and BERTScore (Zhang et al., 2020) against the Chinese reference summaries. ROUGE measures lexical overlap between generated and reference summaries, while BERTScore provides a contextual embedding-based estimate of semantic similarity.

**Learned-metric evaluation.** We additionally use the reference-based OmniScore (Alam et al., 2026) as a learned metric for summary quality and faithfulness. OmniScore evaluates each generated summary relative to the Chinese reference along four dimensions: informativeness, clarity, plausibility, and faithfulness.

| Model          | No err.     | Halluc.     | Omit        | Lang.       |
|----------------|-------------|-------------|-------------|-------------|
| Aya Expanse 8B | 0.38        | 0.43        | 0.07        | 0.17        |
| Gemma 4 E4B    | <b>0.63</b> | <b>0.29</b> | 0.08        | <b>0.00</b> |
| Qwen 3.5 9B    | 0.58        | 0.39        | <b>0.03</b> | <b>0.00</b> |
| mBART-large    | 0.20        | 0.61        | 0.48        | 0.11        |

Table 2: Aggregated grounding-error rates. Values are proportions of the 100 annotated outputs. *Halluc.* includes any H-\* label; error categories may overlap. Outputs with at least one error were 62 for Aya, 37 for Gemma, 42 for Qwen, and 80 for mBART.

## 3 Results

### 3.1 SLMs Exhibit Distinct Dialogue-Semantic Failures

Table 2 summarizes the aggregated error rates. The SLMs produced substantially more error-free summaries and far fewer omissions than mBART, but they still exhibited many source-grounded hallucination errors. Overall, nearly half of the SLM outputs (141/300; 47%) received at least one error label. Because labels may co-occur, category rates should be interpreted as overlapping diagnostic sig-

| Model          | R-1          | R-2          | R-L          | BERTScore    | Info.       | Clar.       | Plaus.      | Faith.      |
|----------------|--------------|--------------|--------------|--------------|-------------|-------------|-------------|-------------|
| Aya Expanse 8B | 26.53        | 7.89         | 20.84        | 70.11        | <b>3.26</b> | 4.02        | 4.04        | <b>3.17</b> |
| Gemma 4 E4B    | 30.27        | 8.32         | 23.68        | 70.85        | 3.17        | 3.95        | 4.00        | 2.96        |
| Qwen 3.5 9B    | 28.35        | 7.62         | 22.42        | 70.47        | 3.22        | 3.88        | 3.97        | 3.00        |
| mBART-large    | <b>41.95</b> | <b>18.10</b> | <b>34.85</b> | <b>77.64</b> | 2.97        | <b>4.24</b> | <b>4.17</b> | 2.92        |

Table 3: Mean automatic-evaluation scores. OmniScore uses its original scale; bold marks the best score for each metric.

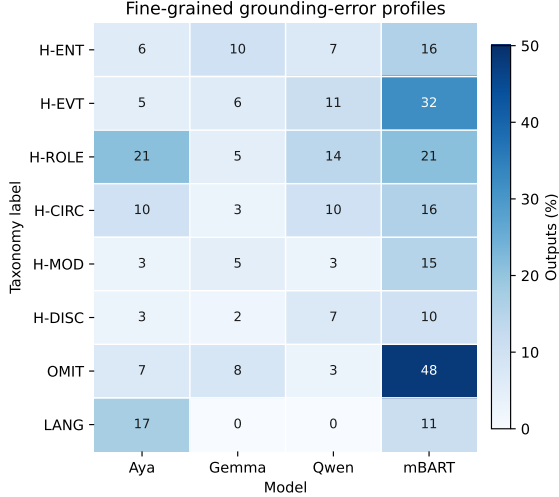


Figure 1: Fine-grained taxonomy-label rates for the three SLMs and the mBART baseline. Each cell reports the percentage of a model’s 100 outputs assigned the corresponding error label. Labels may overlap.

nals rather than mutually exclusive classes.

**Model-specific failure profiles.** Aya had the lowest NO-ERROR rate among the SLMs, with 43 hallucinated outputs, 7 omissions, and 17 language or format errors. Speaker-role hallucination was its dominant error type, appearing in 21 outputs. Most of Aya’s LANG errors involved reproducing translated speaker turns instead of generating a summary. Gemma produced 63 error-free outputs and 29 outputs with hallucinations. Its most frequent error involved entities, and only 2 outputs received multiple labels. Qwen produced 58 error-free outputs and 39 outputs with hallucinations. Its errors were distributed across speaker roles, events, circumstances, and discourse relations. Qwen also produced the fewest omissions but the largest number of multi-label outputs among the three SLMs.

**Taxonomy-specific patterns.** Figure 1 reports the fine-grained error label distribution for all four systems. Among the SLMs, speaker-role hallucination was the most frequent error, appearing in 40 outputs. These errors changed who performed an action, made a statement, offered assistance, or

received an action. Entity and circumstantial hallucinations each appeared in 23 SLM outputs, while event hallucination appeared in 22 outputs. In contrast, mBART showed a different profile, with many omissions and event hallucinations in addition to role errors. Discourse and modality errors were less frequent for the SLMs, appearing in 12 and 11 outputs, respectively, but they also appeared in the mBART baseline.

### 3.2 Automatic Evaluation

Table 3 presents the reference-based evaluation results. The supervised mBART baseline achieved the highest ROUGE and BERTScore values, while Gemma performed best among the zero-shot SLMs, with only small differences across the three SLMs.

The reference-based OmniScore dimensions showed a less uniform pattern. mBART received the highest clarity and plausibility scores but the lowest faithfulness score, whereas Aya obtained the highest faithfulness score among all systems.

Distributional statistics for all automatic metrics are provided in Appendix A.6.

## 4 Discussion

### 4.1 Relation-Sensitive Grounding Failures

The dominant SLM failures were not primarily omissions or output-form failures. As Table 2 shows, omission rates for the SLMs are only 0.03–0.08 and language/format error rates are 0.00–0.17, while hallucination rates are higher at 0.29–0.43. Figure 1 shows that speaker-role hallucination was the most frequent hallucination type, appearing in 40 SLM outputs, while entity, event, and circumstantial errors appeared in 23, 22, and 23 outputs, respectively. These patterns indicate that relation-sensitive distortions are an important failure mode in this setting, particularly for speaker attribution, while the broader taxonomy categories may also include value substitutions or unsupported inventions.

Speaker-role hallucination illustrates this pattern: a summary may mention relevant participants or

events but reverse who performed an action, made a request, offered assistance, or received it. Such errors are especially consequential in dialogue summarization, where requests, decisions, offers, and obligations depend on accurate speaker attribution across turns. Other frequent errors altered participants, events, pragmatic meaning, or contextual details, keeping summaries dialogue-related but structurally misaligned.

Modality and discourse errors were less frequent but still important because they changed whether an event was certain, intended, negated, agreed upon, or concluded. Faithful cross-lingual dialogue summarization therefore requires tracking not only entities and events, but also participant roles, episodic status, and conversational outcomes across turns.

#### 4.2 Model Profiles Suggest Different Failure Modes

The model-level profiles suggest that faithfulness failures arise through different pathways. Aya’s errors were concentrated in speaker-role confusion and output-form problems, including cases where it reproduced translated speaker turns rather than generating a summary. This indicates that multilingual generation ability does not necessarily guarantee task control or successful compression into the required summary format.

Gemma and Qwen showed different SLM profiles. Gemma produced the most error-free outputs, with remaining errors that were relatively localized, often involving entities, omissions, events, or modality. Qwen produced fewer omissions but more distributed and multi-label errors, suggesting that a single misinterpretation can propagate across roles, events, circumstances, and discourse relations. These differences show why a single hallucination rate is insufficient for characterizing SLM reliability.

The mBART baseline exhibited a distinct failure mode. Despite being supervised, it showed many omissions and event hallucinations. This contrast reinforces the value of taxonomy-based analysis: it reveals not only whether a system fails, but how its failures are distributed across dialogue-semantic components.

#### 4.3 Reference-Based Metrics Are Auxiliary to Taxonomy-Based Diagnosis

This contrast is clearest for the mBART baseline. mBART achieved the highest ROUGE-1, ROUGE-

2, ROUGE-L, and BERTScore values, as well as the highest OmniScore clarity and plausibility scores. Under reference-based automatic evaluation alone, it would appear to be the strongest system. However, the manual taxonomy gives the opposite picture for source-grounded faithfulness: mBART had the lowest NO-ERROR rate, the highest hallucination rate, and by far the highest omission rate. This shows that high reference similarity and fluent output form do not guarantee faithful representation of the source dialogue.

A similar divergence appears within the three zero-shot SLMs, which share the same prompt and inference setup. Aya received the highest OmniScore faithfulness score (3.17) among all systems, while also showing the lowest NO-ERROR rate (0.38) and the highest hallucination rate (0.43) among the SLMs. This within-condition discrepancy further suggests that reference-based faithfulness scores need not align with source-grounded error diagnosis.

The discrepancy reflects a limitation of reference-based evaluation. An output can score well against a reference while omitting salient events, decisions, participant relations, or discourse outcomes. Conversely, an output may include source-supported details not selected by the reference, lowering reference similarity without necessarily indicating hallucination. Reference summaries are useful quality anchors, but they cannot directly determine whether a generated summary preserves the dialogue situation. Taxonomy-based diagnosis fills this gap by identifying which part of the dialogue meaning was altered, omitted, or unrealized.

## 5 Conclusion

We presented a taxonomy-based diagnosis of faithfulness failures in zero-shot English-to-Chinese dialogue summarization with SLMs. Their errors included prominent relation-sensitive distortions involving speaker roles, events, circumstances, modality, and discourse outcomes. These findings support source-grounded taxonomy analysis as the primary diagnostic, with reference-based metrics serving as complementary indicators of summary quality.

## 6 Limitations

Our findings have several limitations. We evaluate a fixed 100-dialogue XSAMSum sample in

one English-to-Chinese direction, using one direct prompt and three SLMs under 10B parameters, so the observed label distributions may not generalize across domains, languages, prompts, decoding settings, or model families. Because the taxonomy was induced from preliminary outputs and model-proposed candidate labels were reviewed by three annotators, it may reflect system-specific failure patterns and remain susceptible to anchoring and judgment-sensitive category overlap. Cross-system comparisons are also not controlled performance rankings, since mBART is supervised whereas the SLMs are zero-shot with different decoding configurations. Finally, the use of a single Chinese reference and no English monolingual control limits our ability to separate translation from summarization errors. The taxonomy identifies the affected dialogue-semantic component but does not distinguish rebinding, value falsification, and unanchored invention; accordingly, we do not interpret every H-\* label as a relational error in the strict sense.

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## A Appendix

### A.1 Data

XSAMSum pairs English messenger-style dialogues with abstractive English summaries and professional Chinese and German translations (Wang

et al., 2022b). We use the English dialogues as inputs and the Chinese summaries as references. Table 4 reports the split-level statistics together with the 100-example evaluation subset. The subset is slightly shorter than the full test split in dialogue and reference length, while retaining similar average numbers of turns and speakers.

| Metric            | Train  | Dev   | Test  | Eval. subset |
|-------------------|--------|-------|-------|--------------|
| Dialogues         | 14,732 | 818   | 819   | 100          |
| Turns             | 11.17  | 10.83 | 11.25 | 10.87        |
| Speakers          | 2.39   | 2.39  | 2.36  | 2.43         |
| Dialogue words    | 93.79  | 91.64 | 95.51 | 88.21        |
| En. summary words | 20.32  | 20.28 | 20.02 | 18.62        |
| Zh. summary chars | 35.88  | 35.99 | 35.05 | 33.18        |

Table 4: XSAMSum statistics. Except for the number of dialogues, values are averages. Dialogue and English-summary lengths use whitespace-delimited words; Chinese-summary length counts non-whitespace characters. The evaluation subset is sampled from the test split with seed 42.

### A.2 Model Details

**mBART fine-tuning.** Our supervised baseline is initialized from facebook/mbart-large-50-many-to-many-mmt (Liu et al., 2020; Tang et al., 2021) and fine-tuned to map English XSAMSum dialogues directly to Chinese reference summaries. We use Hugging Face’s Seq2SeqTrainer on a Google Colab A100 40GB GPU, optimizing with Adafactor for four epochs with mixed precision and gradient checkpointing. The best checkpoint is selected by validation ROUGE-Lsum. Source and target tokenizer languages are set to en\_XX and zh\_CN, and generation is forced to start with the Chinese language token. Table 5 summarizes the main settings.

| Setting                  | Value                    |
|--------------------------|--------------------------|
| Maximum input length     | 1024 tokens              |
| Maximum target length    | 128 tokens               |
| Epochs                   | 4                        |
| Learning rate            | $2 \times 10^{-5}$       |
| Per-device batch size    | 4                        |
| Gradient accumulation    | 6 steps                  |
| Effective batch size     | 24                       |
| Optimizer                | Adafactor                |
| Precision                | Mixed precision          |
| Checkpoint selection     | Validation ROUGE-Lsum    |
| Decoding                 | Beam search with 4 beams |
| Source / target language | en_XX / zh_CN            |
| Hardware                 | NVIDIA A100 40GB         |

Table 5: mBART fine-tuning and decoding configuration.

**SLM direct inference.** We evaluate the three instruction-tuned SLMs in the same zero-shot direct setting. Each model receives the English dialogue and generates a Simplified Chinese summary without demonstrations, intermediate translation, or task-specific fine-tuning. The models are run locally through Ollama with a temperature of 0.2, an 8,192-token context window, a maximum of 1,024 generated tokens, and explicit reasoning output disabled.

- **Qwen 3.5 9B.** We use qwen3.5:9b, an instruction-tuned Qwen-family model selected for its multilingual understanding and text-generation capabilities (Qwen Team, 2026).
- **Gemma 4 E4B.** We use gemma4:e4b, an efficient instruction-tuned model from the Gemma family, as a lightweight multilingual baseline (Gemma Team, 2026).
- **Aya Expansive 8B.** We use aya-expansive:8b, a multilingual instruction-tuned model designed to support generation across 23 languages (Dang et al., 2024).

### A.3 Direct Prompt

All three SLMs receive the prompt template in Figure 2, with {dialogue} replaced by the speaker-prefixed English input.

You are an expert English-Chinese bilingual speaker. Given an English dialogue, please write a concise Simplified Chinese summary. Output only the summary, without titles or markdown.

English dialogue:  
{dialogue}

Chinese summary:

Figure 2: Direct prompt template used for zero-shot SLM inference.

### A.4 Annotation Workflow

For the dialogue-semantic grounding analysis, we used a model-assisted first-pass workflow. An annotation model proposed candidate labels for each generated summary, but was not used as an evaluated summarization system or final annotator. Three annotators then reviewed each candidate label against the English source dialogue, Chinese reference summary, generated Chinese summary, and taxonomy definitions in Table 1.

The review prioritized source-grounded faithfulness: hallucination labels were assigned only when the generated summary introduced or altered information relative to the dialogue, while OMIT and LANG captured missing salient content and output-form failures. Multi-label annotations were allowed for independent error types, and uncertain cases were discussed before finalizing labels.

### A.5 Representative Taxonomy Examples

Table 6 presents one representative single-label example for each taxonomy category. We selected short cases with explicit annotation rationales from the analyzed outputs. Generated Chinese summaries are shown as English back-translations for readability.

### A.6 Extended Automatic Evaluation

Table 7 and Table 8 supplement the mean scores in Table 3 with dispersion, median, range, and 95% confidence intervals. Each model contributes 100 outputs.

### A.7 Overall SLM Label and Outcome Distributions

Figure 3 reports the overall occurrence counts for each taxonomy label across the 300 SLM outputs, excluding the mBART baseline. Figure 4 summarizes the corresponding binary semantic outcome. Multi-label outputs contribute once to each assigned label.

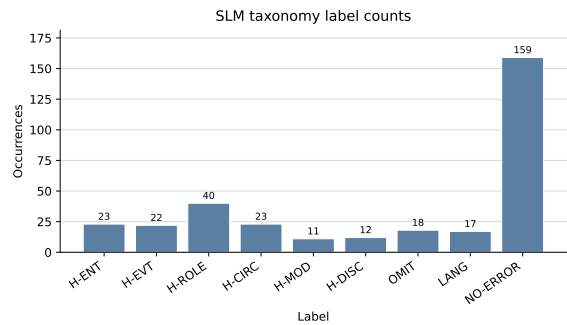


Figure 3: Overall taxonomy-label occurrence counts across the three SLMs, excluding mBART. Multi-label outputs are decomposed into individual label occurrences.

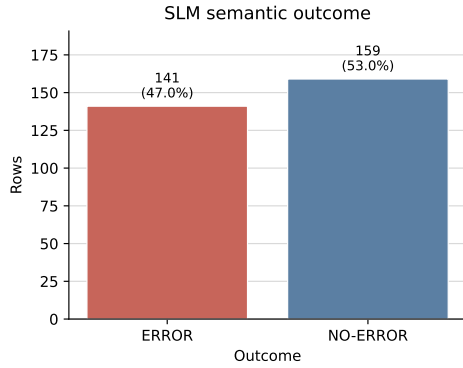


Figure 4: Overall semantic outcome counts across the three SLMs, excluding mBART. Outputs labeled NO-ERROR are counted as no-error cases; all other outputs are counted as error cases.

### A.8 Relation to Prior Dialogue-Summarization Taxonomies

Table 9 summarizes approximate correspondences between our categories and the dialogue-summarization taxonomies of [Tang et al. \(2022\)](#) and [Wang et al. \(2022a\)](#). The categories are not exact equivalents: our scheme is organized by the dialogue-semantic component affected and is applied by comparing each cross-lingual output directly with its source dialogue.

| Label (model, index) | Source dialogue excerpt                                                                               | Generated summary (back-translated)                                                                | Annotation rationale                                                        |
|----------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| H-ENT (Qwen, 30)     | Desiree: "Are you both at home?" Lucian: "No." Desiree: "Keep my pasta in the microwave."             | Debbie asks whether Lucian is home and tells him to leave her pasta in the microwave.              | The summary changes Desiree's identity to Debbie.                           |
| H-EVT (Qwen, 21)     | Adrian: "Can you talk?" Simon: "Not really. I'll be free at 5."                                       | They arrange to meet and talk at five.                                                             | A later conversation is changed into an in-person meeting.                  |
| H-ROLE (mBART, 43)   | Daisy: "Going home?" Lisa: "Not yet." Daisy: "Please be back before 11 pm."                           | Daisy wants to be home before 11 p.m.; Lisa is not home yet.                                       | The return requirement is assigned to Daisy instead of Lisa.                |
| H-CIRC (Gemma, 3)    | Mary: "Lend me a few bucks." Carter: "Give me an hour; I'm at the train station."                     | Carter will give Mary the money at the train station in one hour.                                  | Carter's current location becomes the future handover location.             |
| H-MOD (Gemma, 60)    | John: "Do we have practice today?" Matt: "Yeah, think so."                                            | Matt confirms that they have practice today.                                                       | The uncertain "think so" becomes a definite confirmation.                   |
| H-DISC (Qwen, 22)    | Gab asks Kat to meet for a drink. Kat says she is busy and does not meet strangers from the internet. | Gab and Kat agree to meet, but Kat later refuses because of the risks of meeting online strangers. | The summary fabricates an initial agreement and subsequent reversal.        |
| OMIT (Gemma, 15)     | Pete: "Did you clean the hamster cage?" Adelle: "No." Pete: "After school, no excuses."               | Pete tells Adelle to clean the hamster cage, and she agrees.                                       | The salient after-school timing is omitted.                                 |
| LANG (Aya, 77)       | Dolph: "Remember to bring your passport tomorrow." Ziggler: "Cool, I will."                           | Dolph: "Remember to bring your passport tomorrow." Ziggler: "Okay, I will remember."               | The output translates and reproduces the turns instead of summarizing them. |
| NO-ERROR (Gemma, 43) | Daisy: "Going home?" Lisa: "Not yet." Daisy: "Please be back before 11 pm."                           | Daisy asks Lisa to be home before 11 p.m.                                                          | The key event, participant roles, and timing are preserved.                 |

Table 6: Representative examples from the annotated outputs. Parentheses identify the model and sample index. Generated Chinese summaries are shown as English back-translations; dialogue excerpts are shortened without changing the evidence relevant to the assigned label.

| Model | Metric    | SD    | Median | Min-Max      | 95% CI      |
|-------|-----------|-------|--------|--------------|-------------|
| Aya   | ROUGE-1   | 10.58 | 24.28  | 5.26-51.28   | 24.46-28.60 |
| Aya   | ROUGE-2   | 6.81  | 5.97   | 0.00-28.28   | 6.55-9.22   |
| Aya   | ROUGE-L   | 9.02  | 19.23  | 5.26-51.28   | 19.07-22.60 |
| Aya   | BERTScore | 4.60  | 70.33  | 59.70-83.88  | 69.21-71.02 |
| Gemma | ROUGE-1   | 11.35 | 30.28  | 0.00-61.54   | 28.05-32.50 |
| Gemma | ROUGE-2   | 7.51  | 7.37   | 0.00-34.15   | 6.85-9.79   |
| Gemma | ROUGE-L   | 10.30 | 24.32  | 0.00-54.55   | 21.66-25.70 |
| Gemma | BERTScore | 5.58  | 71.06  | 57.16-84.79  | 69.76-71.94 |
| Qwen  | ROUGE-1   | 11.12 | 27.09  | 3.88-53.85   | 26.17-30.53 |
| Qwen  | ROUGE-2   | 8.27  | 5.27   | 0.00-48.48   | 6.00-9.24   |
| Qwen  | ROUGE-L   | 10.03 | 21.05  | 3.88-51.95   | 20.45-24.38 |
| Qwen  | BERTScore | 4.72  | 70.32  | 56.95-83.35  | 69.55-71.40 |
| mBART | ROUGE-1   | 19.06 | 37.97  | 8.33-100.00  | 38.21-45.68 |
| mBART | ROUGE-2   | 20.10 | 12.50  | 0.00-100.00  | 14.16-22.04 |
| mBART | ROUGE-L   | 18.62 | 31.17  | 8.33-100.00  | 31.21-38.50 |
| mBART | BERTScore | 8.40  | 76.16  | 60.94-100.00 | 76.00-79.29 |

Table 7: Distributional statistics for ROUGE and BERTScore. Mean values are reported in Table 3.

| Model | OmniScore dimension | SD   | Median | Min-Max   | 95% CI    |
|-------|---------------------|------|--------|-----------|-----------|
| Aya   | Informativeness     | 0.58 | 3.31   | 2.00-4.74 | 3.15-3.38 |
| Aya   | Clarity             | 0.37 | 3.99   | 2.95-4.93 | 3.95-4.09 |
| Aya   | Plausibility        | 0.37 | 4.02   | 2.86-4.93 | 3.97-4.12 |
| Aya   | Faithfulness        | 0.51 | 3.09   | 2.15-4.74 | 3.07-3.27 |
| Gemma | Informativeness     | 0.56 | 3.13   | 2.01-4.66 | 3.06-3.28 |
| Gemma | Clarity             | 0.32 | 3.94   | 2.87-4.93 | 3.88-4.01 |
| Gemma | Plausibility        | 0.31 | 3.95   | 2.89-4.93 | 3.94-4.06 |
| Gemma | Faithfulness        | 0.42 | 2.96   | 2.03-3.87 | 2.88-3.04 |
| Qwen  | Informativeness     | 0.50 | 3.30   | 1.87-4.31 | 3.12-3.32 |
| Qwen  | Clarity             | 0.26 | 3.90   | 3.17-4.72 | 3.83-3.93 |
| Qwen  | Plausibility        | 0.26 | 3.97   | 3.09-4.77 | 3.92-4.02 |
| Qwen  | Faithfulness        | 0.34 | 3.01   | 2.01-3.73 | 2.94-3.07 |
| mBART | Informativeness     | 0.68 | 2.85   | 1.53-4.84 | 2.83-3.10 |
| mBART | Clarity             | 0.41 | 4.22   | 3.36-4.93 | 4.16-4.32 |
| mBART | Plausibility        | 0.42 | 4.13   | 3.35-4.94 | 4.08-4.25 |
| mBART | Faithfulness        | 0.66 | 2.79   | 1.65-4.74 | 2.79-3.05 |

Table 8: Distributional statistics for the four reference-based OmniScore dimensions. Mean values are reported in Table 3.

| Our category | Tang et al. (2022)          | Wang et al. (2022)       | Primary emphasis in our taxonomy                             |
|--------------|-----------------------------|--------------------------|--------------------------------------------------------------|
| H-ENT        | Object / Wrong Reference    | Subject-Object / Pronoun | Entity identity, separated from event and participant role   |
| H-EVT        | —                           | Hallucination            | Added, changed, or fabricated event                          |
| H-ROLE       | Wrong Reference             | Subject-Object           | Actor-recipient-speaker relation                             |
| H-CIRC       | Circumstantial              | Particulars              | Time, place, quantity, condition, or schedule                |
| H-MOD        | Negation / Tense / Modality | Negation                 | Negation, uncertainty, intention, obligation, or modal force |
| H-DISC       | —                           | Other                    | Cause, contrast, order, decision, agreement, or outcome      |
| OMIT         | Missing Information         | —                        | Salient information absent from the output                   |
| LANG         | —                           | —                        | Cross-lingual language, format, or task-fulfillment failure  |

Table 9: Approximate correspondence between our taxonomy and prior factual-error taxonomies for dialogue summarization. A dash indicates that the prior work has no directly corresponding standalone category; it does not imply that the broader phenomenon is entirely absent.